UNIVERSITY OF NAIROBI — FACULTY OF SCIENCE & TECHNOLOGY
DEPARTMENT OF COMPUTING & INFORMATICS | CHIROMO CAMPUS

UON CHIROMO YEAR 2: HIGH-YIELD FORMULA & EXAM CRAM SHEET HANDBOOK

Condensed Revision Sheets, Algorithm Tracing Guides, Core Theorems, and Examiner Trap Warnings for All 7 Year 2 Units

Curriculum Year & Scope:	**Bachelor of Science in Computer Science — Year 2 (Full Scope)**
Academic Units Covered:	• SCS 3201: Systems Analysis & Design • SCS 3203: Computer Architecture • SCS 3205: Knowledge-Based Systems • SCS 3207: Operating Systems • SCS 3209: Software Engineering • CSC 3211: Computer Networks • SMA 2114: Differential & Integral Calculus
Examinations Solved:	UoN Semester Examinations & CATs (2018, 2019, 2024, 2025, 2026)
Primary Prescribed Texts:	• Tanenbaum (OS 4th ed.) | Stallings (OS 9th ed.) • Stallings (Comp Arch 10th ed.) | Patterson & Hennessy (MIPS) • Russell & Norvig (AIMA 3rd ed.) | Akerkar (KBS) • Forouzan (Networks 4th ed.) | Comer (Networks 6th ed.) • Kendall & Kendall (SAD 10th ed.) | Sommerville (SE 9th/10th ed.)
Student Author:	Asman Makero Musa (P15/XXXXX/2024)
Offline Readiness:	**100% SELF-CONTAINED — NO INTERNET CONNECTION REQUIRED**

UNIT 1: SCS 3201 — SYSTEMS ANALYSIS AND DESIGN (CRAM SHEET)

1. DFD Rules (Kendall & Kendall Ch. 7):
• **Black Hole Trap:** Process has inputs but zero outputs (Data enters and disappears).
• **Miracle Trap:** Process produces outputs with zero inputs.
• **Gray Hole Trap:** Inputs are insufficient to produce the asserted outputs.
• **Direct Flow Prohibitions:** Data cannot flow directly between External Entity ↔ External Entity, nor Data Store ↔ Data Store, nor External Entity ↔ Data Store. Every flow MUST pass through a Process.

2. TELOS Feasibility: T = Technical, E = Economic (ROI, Payback), L = Legal, O = Operational, S = Schedule.

3. Normalization Rules (1NF, 2NF, 3NF):
• **1NF:** Atomic values only; no repeating groups.
• **2NF:** In 1NF + No partial dependencies (all non-key attributes fully functionally dependent on whole composite primary key).
• **3NF:** In 2NF + No transitive dependencies (no non-key attribute depends on another non-key attribute: A → B and B → C).

UNIT 2: SCS 3203 — COMPUTER ARCHITECTURE (CRAM SHEET)

1. IEEE-754 32-bit Single Precision Format:
• 1 Sign bit (s) | 8 Exponent bits (E) with Bias = 127 | 23 Fraction/Mantissa bits (F)
• Value = (-1)^s · (1.F) · 2^(E - 127). Special cases: E=255, F=0 → ±∞; E=255, F≠0 → NaN.

2. Cache Address Calculation (Memory = 2^m bytes, Cache = 2^C bytes, Block = 2^b bytes):
• Word Offset = b bits.
• Direct Mapped: Number of lines N = 2^(C-b). Line Index = C-b bits. Tag = m - C bits.
• k-way Set Associative: Sets S = N / k = 2^(C-b) / k. Set Index = log2(S) bits. Tag = m - log2(S) - b bits.

3. Pipelining Hazards:
• **Structural:** Hardware resource conflict (e.g., single memory port for Instruction and Data).
• **Data:** RAW (Read After Write - true dependency), WAR (anti), WAW (output). Solution: Forwarding/Bypassing or stalls.
• **Control:** Branch penalty. Solution: Dynamic branch prediction (2-bit saturating counter).

UNIT 3: SCS 3207 — OPERATING SYSTEMS (CRAM SHEET)

1. CPU Scheduling Metrics:
• Turnaround Time = Completion Time - Arrival Time.
• Waiting Time = Turnaround Time - Burst Time.
• Response Time = Time from arrival to first CPU allocation.

2. 4 Coffman Deadlock Conditions (ALL 4 must hold simultaneously):
1. Mutual Exclusion (non-shareable resources).
2. Hold and Wait (processes holding resources request new ones).
3. No Preemption (resources cannot be forcibly taken).
4. Circular Wait (P0 waiting for P1, P1 for P2... Pn for P0).

3. Page Replacement Algorithms:
• **FIFO:** Suffers from Belady's Anomaly (increasing frames can increase page faults!).
• **LRU:** Replaces page unreferenced for longest time. Stack algorithm (immune to Belady's Anomaly).
• **Optimal (OPT/MIN):** Replaces page not needed for longest future time (theoretical benchmark).

UNIT 4: CSC 3211 — COMPUTER NETWORKS (CRAM SHEET)

1. Subnetting & CIDR Calculation Cheat Formulas:
• Subnet Mask /N means N network/subnet bits and (32 - N) host bits.
• Total addresses per subnet = 2^(32 - N).
• Usable hosts per subnet = 2^(32 - N) - 2 (subtract Network Address and Broadcast Address).
• Block Size = 256 - Octet Value (e.g., /26 is 255.255.255.192 → Block size = 256 - 192 = 64).

2. OSI 7 Layers vs. TCP/IP 4 Layers:
• 7: Application | 6: Presentation | 5: Session → **TCP/IP Application Layer** (HTTP, DNS, SSH)
• 4: Transport (TCP, UDP) → Segments (TCP provides reliable byte-stream, 3-way handshake SYN, SYN-ACK, ACK)
• 3: Network (IP, ICMP, OSPF, BGP) → Packets (logical addressing, routing)
• 2: Data Link (Ethernet, Wi-Fi, MAC) → Frames (physical addressing, framing, CRC error detection)
• 1: Physical (Bits, Voltage, Fiber) → Bits

UNIT 5: SCS 3205 — KNOWLEDGE-BASED SYSTEMS (CRAM SHEET)

1. Expert System Architecture: User Interface ↔ Inference Engine (Forward/Backward Chaining) ↔ Knowledge Base (Production Rules: IF condition THEN action) + Working Memory (Current facts).

2. Inference Mechanisms:
• **Forward Chaining (Data-Driven):** Starts with known facts in working memory; applies rules to derive new facts until goal reached (used in monitoring, diagnosis, synthesis).
• **Backward Chaining (Goal-Driven):** Starts with hypothesized goal; searches rules that conclude the goal; recurses on preconditions (used in verification, diagnostic query systems, Prolog).

UNIT 6: SCS 3209 — SOFTWARE ENGINEERING (CRAM SHEET)

1. Testing Levels & V-Model:
• Unit Testing (Functions/Classes) ↔ Detailed Design (LLD)
• Integration Testing (APIs/Databases) ↔ Architecture (SAD/HLD)
• System Testing (End-to-End Requirements) ↔ System Requirements (SRS)
• Acceptance Testing (UAT) ↔ User Requirements (URD/BRD)

2. Clean Architecture & GoF Patterns:
• **SOLID Principles:** Single Responsibility, Open/Closed, Liskov Substitution, Interface Segregation, Dependency Inversion.
• **Repository Pattern:** Isolates business logic from database schema details.

UNIT 7: SMA 2114 — DIFFERENTIAL & INTEGRAL CALCULUS (CRAM SHEET)

1. Essential Integration Formulas:
• $\int 1 / (a^2 + x^2)\, dx = (1/a) \arctan(x/a) + C$
• $\int 1 / \sqrt{(a^2 - x^2)}\, dx = \arcsin(x/a) + C$
• $\int \tan(x)\, dx = \ln|\sec(x)| + C$ | $\int \sec(x)\, dx = \ln|\sec(x) + \tan(x)| + C$

2. Trigonometric Substitutions:
• $\sqrt{(a^2 - x^2)} \rightarrow$ Let $x = a \sin(\theta)$, $dx = a \cos(\theta)\, d\theta$
• $\sqrt{(a^2 + x^2)} \rightarrow$ Let $x = a \tan(\theta)$, $dx = a \sec^2(\theta)\, d\theta$
• $\sqrt{(x^2 - a^2)} \rightarrow$ Let $x = a \sec(\theta)$, $dx = a \sec(\theta) \tan(\theta)\, d\theta$